A molecular taxonomy of tumors independent of tissue-of-origin

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Introduction

- Due to decline in cost for next generation sequencing, datasets have become available that allow researchers to understand the genomic landscape of cancer.
- Efforts to distinguish overarching mutational processes from interactions with cell of origin have seen limited success,.
- Thus, there is a need to shift from an overly-simplified traditional model for cancer therapy, a one size fits all, to a precision and personalized medicine approach.

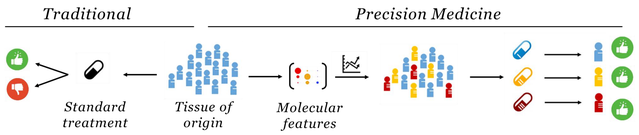

Figure 1: Illustration of the traditional model for cancer treatment (left) and the benefits of precision medicine based on molecular features (right).

Objective

To understand how tissue-specific gene disruption create common cancer phenotypes by focusing on discrete molecular pathways as the unit of disruption

Materials & Methods

Whole-exome sequencing of somatic mutation in the coding region data for 7,607 tumor samples representing 23 cancers from The Cancer Genome Atlas (TCGA).

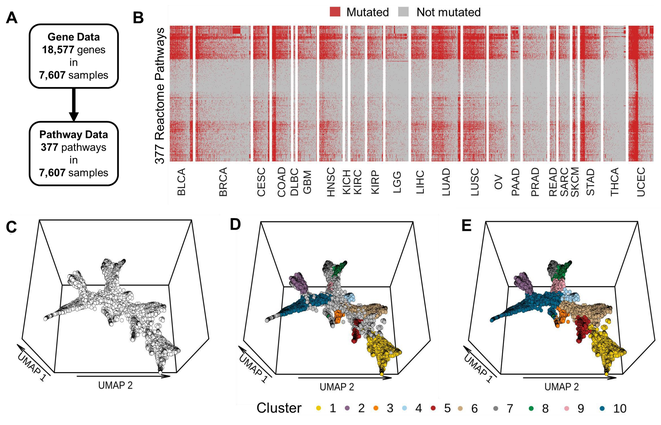

Figure 2: Bioinformatic framework to classify tumor samples based on dysregulation in molecular pathways. A) Gene somatic mutations were transformed to pathway aberration. B) The heatmap shows the molecular pathway mutational profiles (rows) for each tumor samples (columns) sorted by cancer type. C) Results from MCA of the data was projected down to 3 dimensions (UMAP) for the purpose of density-based clustering. Each dot represents a tumor sample. D) UMAP projection of cluster membership extracted from HDBSCAN. Unclassified tumors are colored grey. E) kNN was used to assign membership to unclassified tumors.

Results

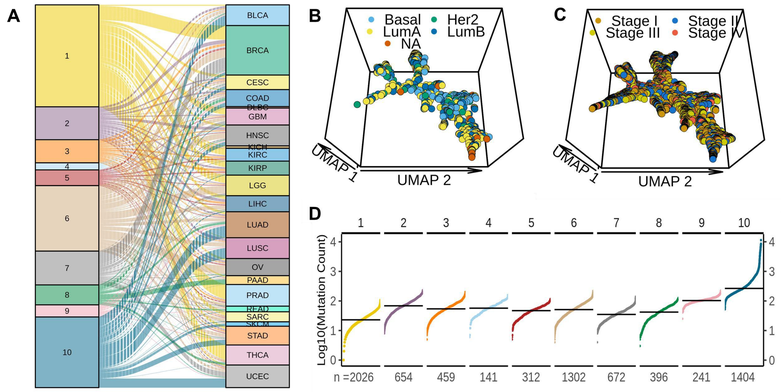

Figure 3: Pathway-based clustering independent of tissue-of-origin. A) Sankey plot of correspondence between cancer type and cluster identity. B) UMAP projection of breast cancer subtypes C) UMAP projection of tumor stage. D) Ranked somatic mutation frequencies for each cluster. Each dot represents a tumor sample. The median mutation count in each cluster is indicated by the horizontal line.

Pathway	M cluster	Effect Size	log10(p)	NM cluster
Signaling by NOTCH1	2	0.34	2.11	
Signaling by NOTCH2	2	0.40	2.82	
Plasma lipoprotein clearance	2	0.31	1.94	
Hedgehog 'off' state	3	0.31	2.07	2
Nonhomologous End-Joining (NHEJ)	3	0.33	2.53	4
Effects of PIP2 hydrolysis	4	0.49	2.04	
Signaling by SCF-KIT	4	0.54	2.55	7, 8, 9
Respiratory electron transport, ATP synthesis by chemiosmotic coupling, and heat production by uncoupling proteins.	4	0.60	2.41	
Fc epsilon receptor (FCERI) signaling	4	0.55	2.12	2, 7, 8
Platelet calcium homeostasis	4	0.44	1.67	
Energy dependent regulation of mTOR by LKB1-AMPK	5	0.31	1.80	
GPCR ligand binding	5	0.35	1.93	
Nonhomologous End-Joining (NHEJ)	5	0.59	3.70	4
DNA Double Strand Break Response	5	0.34	1.63	4, 6
RNA Polymerase II Transcription Pre-Initiation And Promoter Opening	5	0.35	2.36	
Response to elevated platelet cytosolic Ca2+	5	0.34	1.61	
RNA Polymerase II Transcription Initiation And Promoter Clearance	5	0.35	2.38	
Protein ubiquitination	5	0.35	1.87	
Glycogen metabolism	5	0.36	2.52	
TCF dependent signaling in response to WNT	7	0.34	2.90	2, 4
Neurexins and neuroligins	7	0.30	2.71	
Signaling by PTK6	7	0.37	3.21	8
Erythropoietin activates RAS	7	0.31	3.28	8
Pre-NOTCH Expression and Processing	9	0.30	5.00	4, 6
Transcriptional activation of mitochondrial biogenesis	9	0.35	1.80	
Beta-catenin independent WNT signaling	9	0.33	5.00	2, 8
Hedgehog 'off' state	9	0.35	1.83	2
G1/S DNA Damage Checkpoints	9	0.40	5.00	2, 6
Signaling by PTK6	9	0.30	5.00	8
Erythropoietin activates RAS	9	0.38	2.37	8
Butyrate Response Factor 1 (BRF1) binds and destabilizes mRNA	10	0.30	4.00	

Table 1: Cluster-specific enriched pathways (effect size ≥ 0.30) in metastases. M cluster: Enriched in cluster-specific metastatic tumor; NM cluster: Enriched in cluster-specific nonmetastatic tumor.

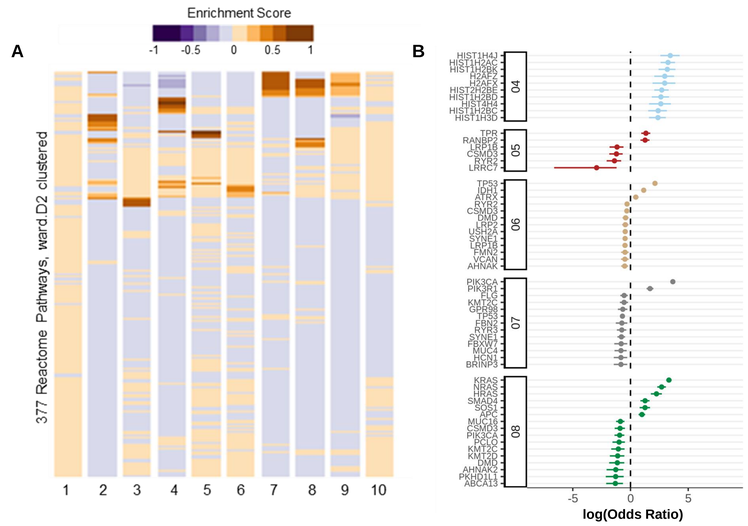

Figure 4: Pan-cancer enrichment of pathway disruptions. A) Heatmap shows cluster enrichment of each pathway relative to other clusters. Color scale indicates effect size (percent enrichment). B) Odds ratio plot with a 95% confidence interval. Cluster is compared to background (all other clusters) to identify the top 10 differentially mutated genes (enriched or depleted).

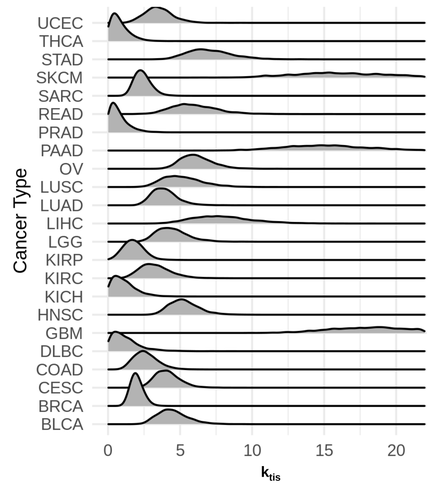

Figure 5: Tissue-specific k represent relative aggressiveness for each cancer type. Larger values correspond to decreased age-independent life expectancy.

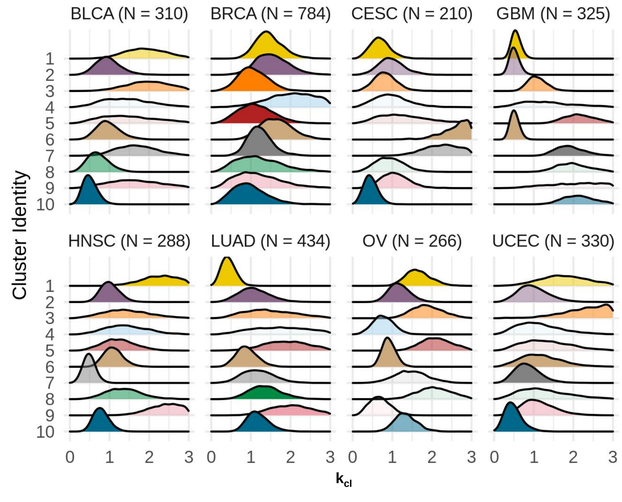

Figure 6: Posterior estimates of cluster-specific k: relative aggressiveness. Alpha transparency is set to reflect varying confidence in the posterior distribution when the number of observations is N<10, α = 0.1; N<25, α = 0.5, N≥25, α = 1.0.

Conclusions

Here, we present a novel bioinformatic analysis to identify the molecular pathways that are commonly mutated between and within different cancers.

- Cancer can be subdivided into ten distinguishable clusters (Fig. 2E) that represent phenotypic classes (Fig. 4) that cut across tissue-of-origin (Fig. 3)
- Cluster-specific enrichment in metastatic tumors is also found in non-metastatic enrichment of other clusters (Table 1).
- By modeling the CDC longevity data as a baseline risk function, we showed that each cluster exhibits cancer type specific effects on survival expectancy (Fig. 5 & 6)

Future Directions

- Integration of other OMICs into the pathway analysis
- Pharmacogenomics studies using identified clusters.

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Interactive media

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